

Intro to GitHub

What is Git?

Git is a free and open-source distributed version control system (DVCS) designed to manage and track changes in source code and other files during development. - A.K.A... it's really good for coding collaboratively & simultaneously— think of it like Google Drive/Google Docs for code.

GitHub.com vs. GitHub Desktop

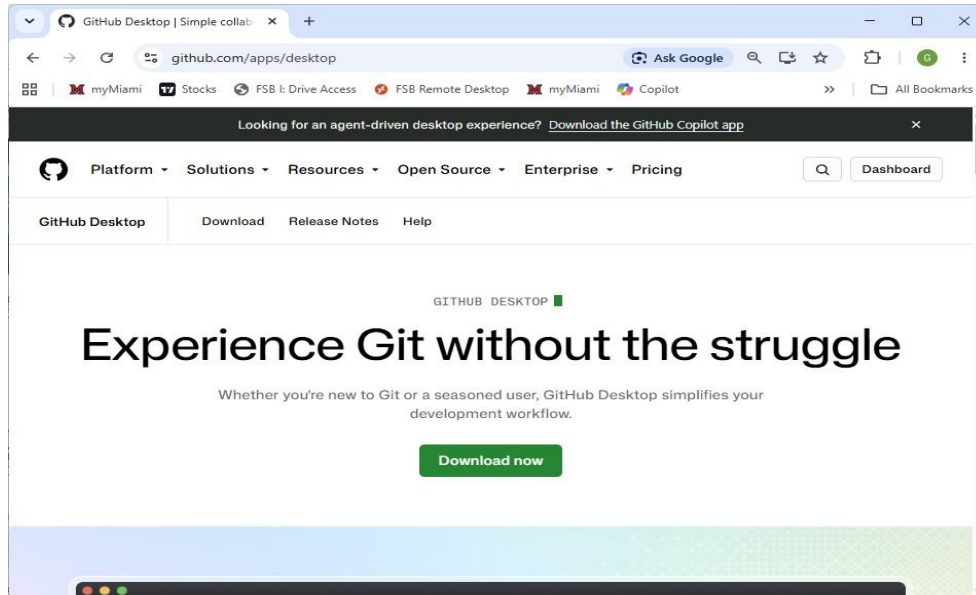
GitHub.com hosts your website server, while the GitHub Desktop app allows you to work on files, push, and pull a lot more easily.

Vocabulary:

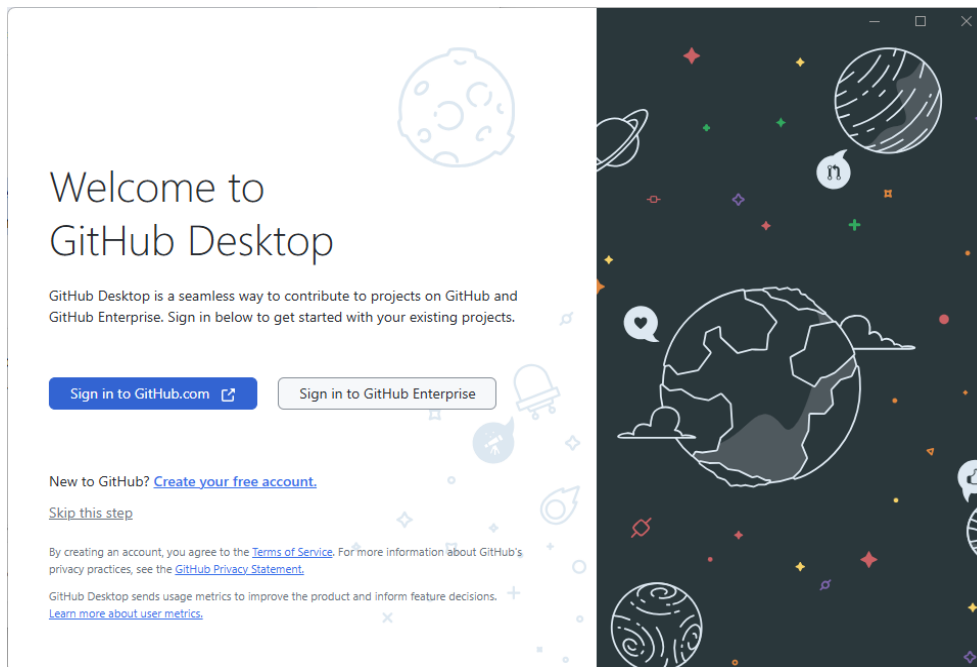
- **Repository** (repo): the location where all your files are stored for a project (think of it like a Google Drive folder)
- **Branch**: a contained line of development within the repository, called main. Don't worry too much about what a branch actually is, because you won't need to create multiple branches in your project.
- **Clone**: one person will create the repository on GitHub.com (the server), and everyone else will have to clone it to have it available on their device.
- **Commit**: Save your changes on your clone of the repository.
- **Push**: Send out your changes to the group (the server).
- **Fetch origin**: Check for any new changes from the server.
- **Pull**: Retrieve other group members' changes and add them to your clone. **ALWAYS PULL BEFORE YOU PUSH!**
- **Merge**: Combine your changes with your group members' changes. Merge conflicts will emerge if you and a group member worked on the same line of code with different results— usually Git will handle these, but sometimes you will have to resolve them manually.

Download Instructions

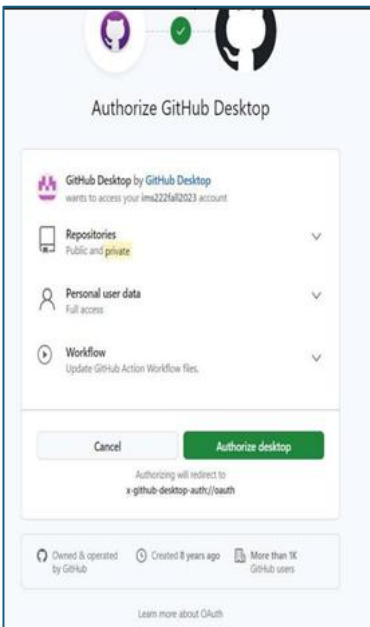
1. Download GitHub Desktop at: <https://github.com/apps/desktop>



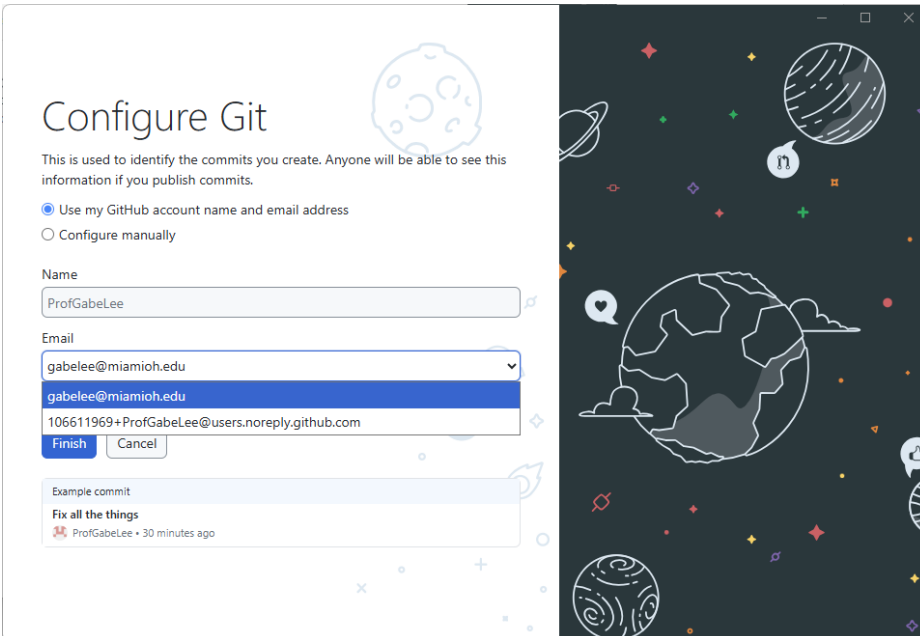
2. Install it. When installed, the app will ask you to sign into your GitHub account.



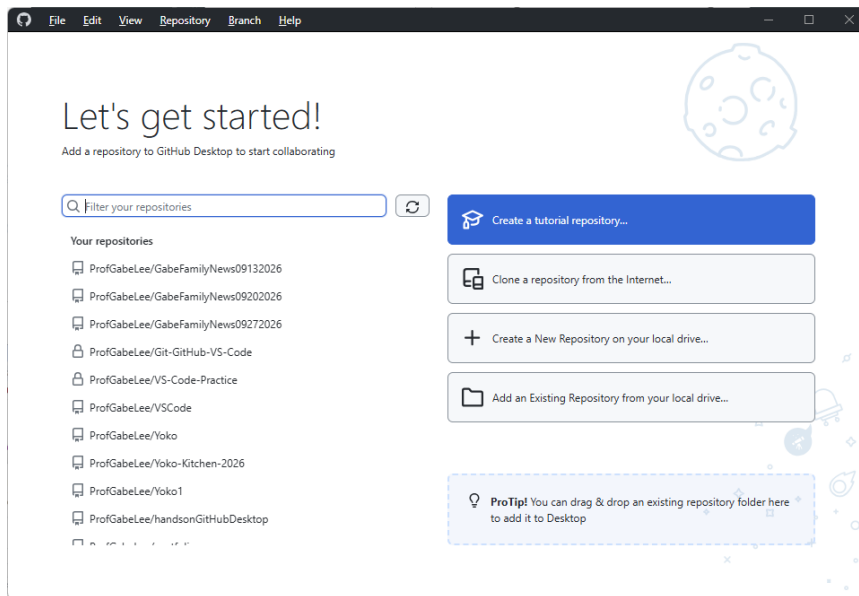
3. Click Authorize and Continue



4. Configure Git and click Finish



5. First look at the Desktop app (without active repository)

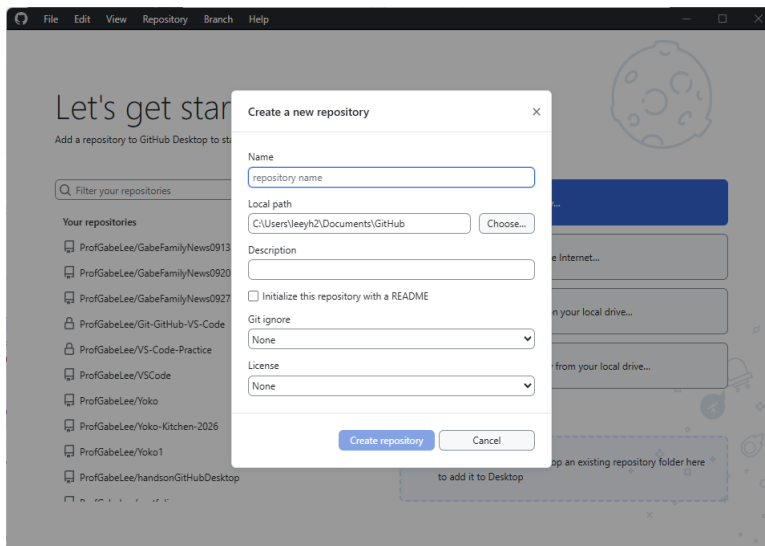


Set Up Your Group's Repository

1. Get in your groups! **Only *one* person will need to create the repository** (everyone else will clone it), so decide who that is going to be.

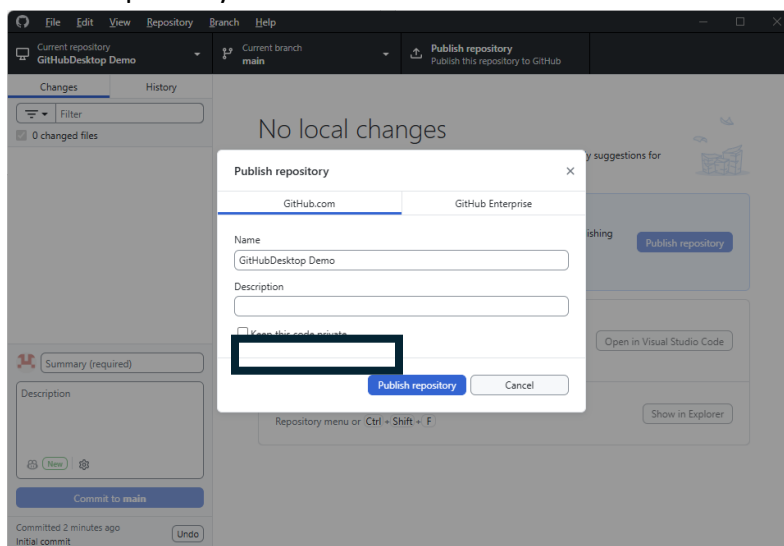
Instructions for ONE person creating the repository

2. Click the “Create a New Repository on your hard drive” and fill out the fields.

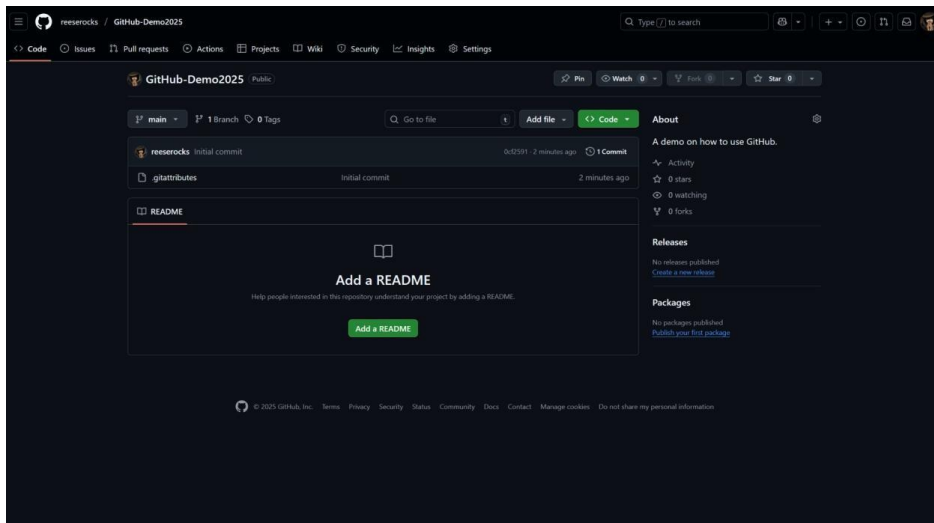


***Be mindful of the local path.** Make sure you know where this repository is within your computer's File Explorer (Windows) and Finder (Mac).

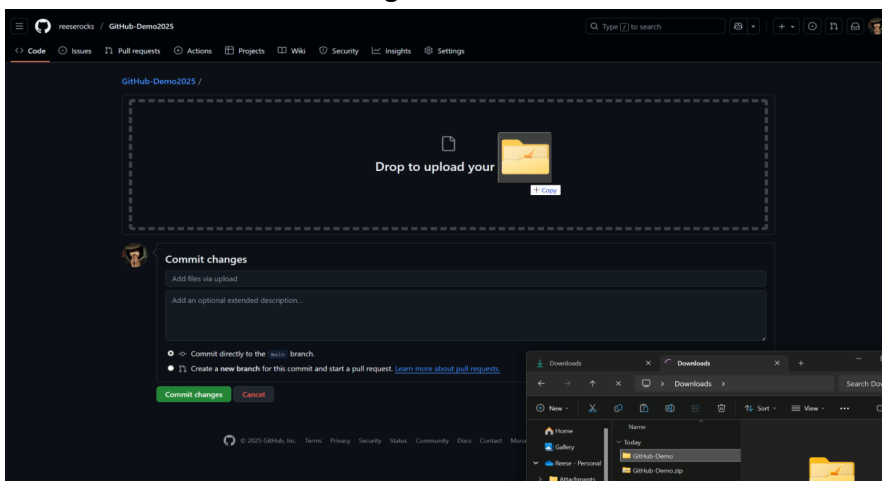
3. Click **Create Repository** button. Then, Click **Publish repository** button. Then, **Uncheck “keep this code private”** on the pop-up and click Publish repository. You may need to add the name of the repository.



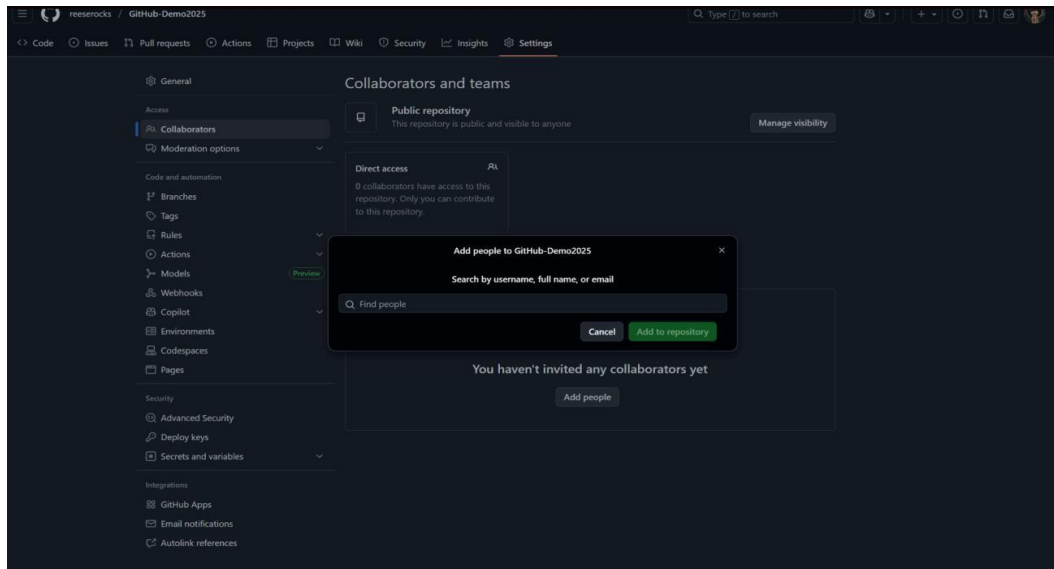
4. Download **GitHubDemo.zip** from Canvas and unzip it.
5. Now, go to <https://github.com>, open up your repositories, and find the repository you just created.



6. Click “Add file” > “Upload Files” on the github.com repository and add the **GitHubDemo** folder. Allow these to finish uploading. Go to **Commit changes** and add “v1” and then click “Commit Changes”

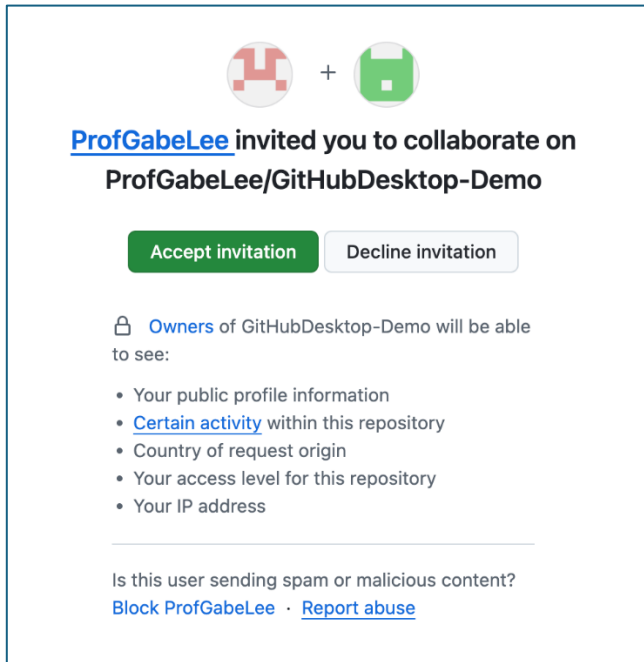


7. Now, you're going to invite your team members so that they can clone this repository. Go to Settings > Collaborators > (enter your password) > Manage Access > Add people. **You add people to GitHub repositories by entering their usernames, so allow your group members to type in their usernames (or email address) now.**

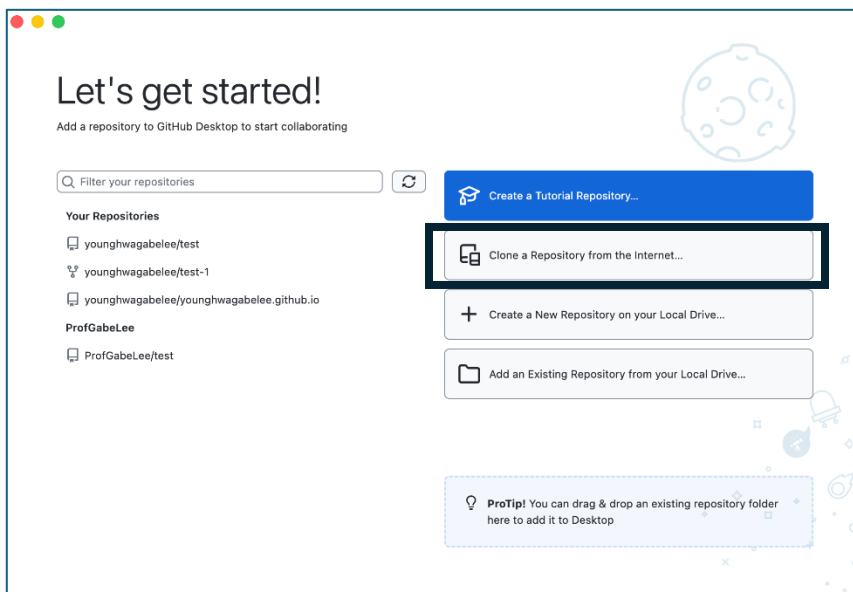


Instructions for everyone else CLONING the repository

8. Each member will go to your email and look for the invitation from GitHub. Then, accept the invitation.

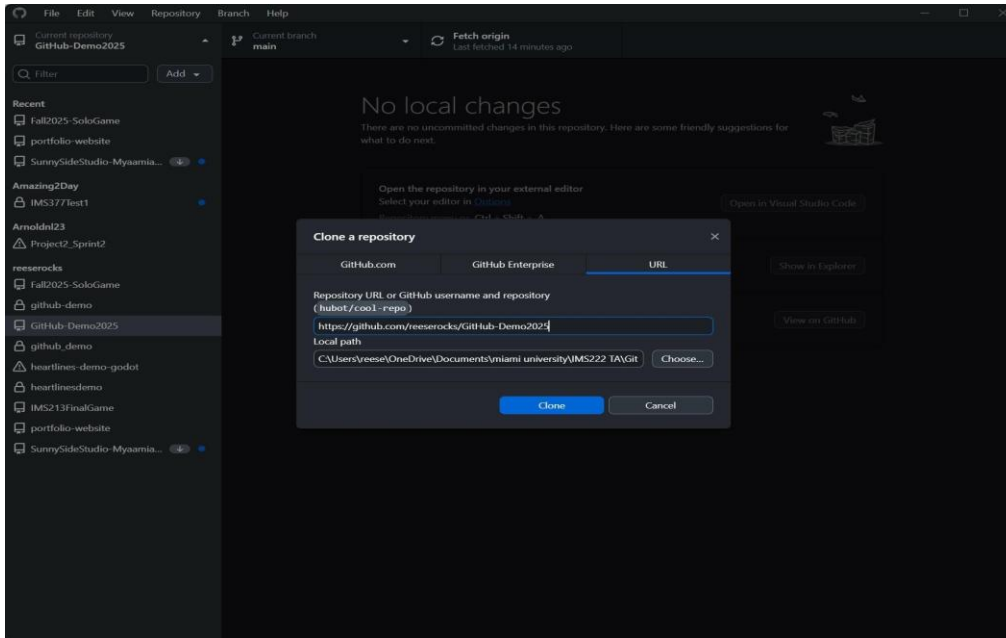


9. With GitHub Desktop open, everyone else should click **“Clone a Repository from the Internet.”**



10. This will add the repository to your individual devices. Now that you've been added as a collaborator, you can find it from a list of your repositories, or you can add it by URL.

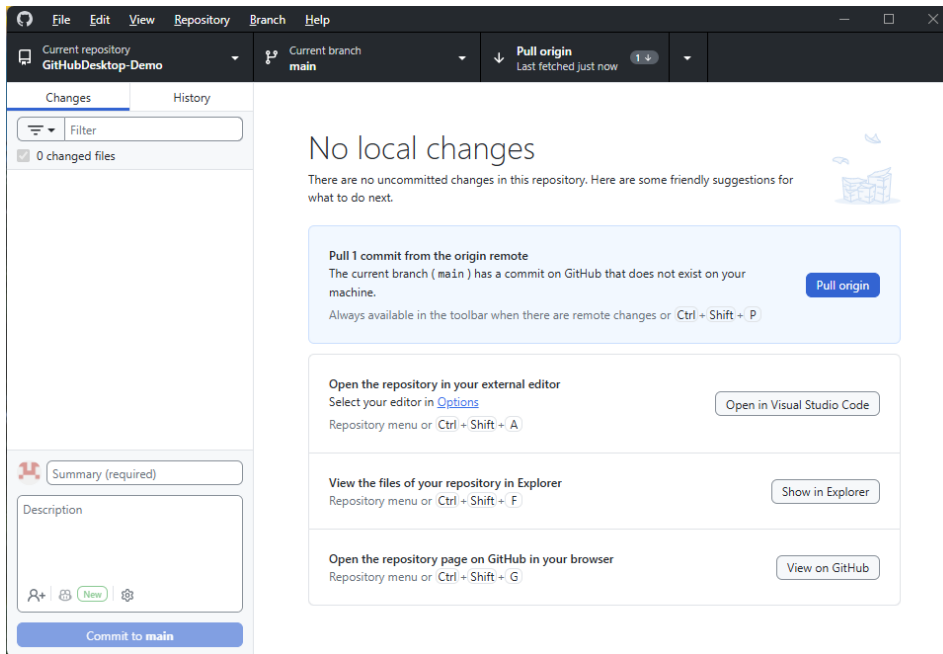
*Again, **be mindful of the local path** so that you know where the repository will be located on your computer.



* Make sure to **fetch origin and pull any changes**, if needed, to ensure all the GitHubDemo files are on your computer now.

Group Hands-On Activity

Everyone, open up Phoenix Code on your computers and open your repository folder. GitHub Desktop is now working in the background to track any changes you make to these files. Unlike Google Docs, you will not see other people's changes as they're working on the files— you will have to **pull** in order to see them.



1. Now, everyone will split up and create content underneath the commented headings, including:
 - index.html:
 - Article 1
 - Article 2
 - List
 - ingredients.html
 - Table
 - Blurb & Image

Be mindful not to work on the same exact lines of code in order to avoid any bad merge conflicts.

Index.html

<!--add article 1 by team member 1-->

```
<article>
  <figure>
    
    <figcaption>Pizza</figcaption>
  </figure>
  <h2>Pizza</h2>
  <p>Pizza usually comes with cheese, sauce, crust, and toppings.</p>
</article>
```

<!--add article 2 by team member 2-->

```
<article>
  <figure>
    
    <figcaption>Pasta</figcaption>
  </figure>
  <h2>Pasta</h2>
  <p>A delicious dish you can make a thousand ways, usually with red or
white sauce.</p> </article>
```

<!--add list by team member 3-->

```
<ul>
  <li>Buffalo Chicken Dip</li> <li>Teriyaki Chicken</li>
  <li>BBQ Braised Short Ribs</li>
  <li>Shrimp Scampi</li>
</ul>
```

Ingredients.html

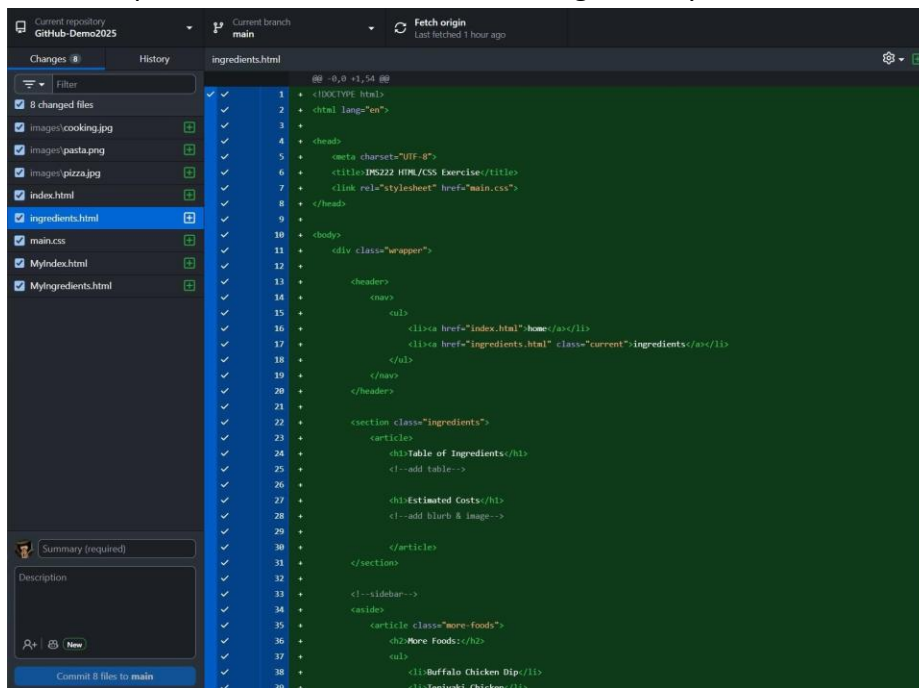
<!--add table by team member 4-->

```
<table>
  <tr>
    <td>Pizza dough</td>
    <td>Tomato sauce</td>
    <td>Mozzarella</td> <td>Olive oil</td>
    <td>Fresh basil</td>
  </tr>
  <tr>
    <td>Spaghetti</td>
    <td>Parmesan</td> <td>Garlic</td>
    <td>Crushed tomatoes</td> <td>Ground beef</td>
  </tr>
  <tr>
    <td>Shrimp</td> <td>Butter</td>
    <td>White wine</td>
    <td>Lemon juice</td> <td>Parsley</td>
  </tr>
  <tr>
    <td>Chicken breast</td>
    <td>Hot sauce</td>
    <td>Ranch dressing</td> <td>Cream cheese</td><td>Cheddar cheese</td>
  </tr>
  <tr>
    <td>Chicken thighs</td>
    <td>Soy sauce</td>
    <td>Mirin</td> <td>Brown sugar</td><td>Ginger</td>
  </tr>
  <tr>
    <td>Beef short ribs</td>
    <td>Red wine</td><td>Beef broth</td>
    <td>Carrots</td><td>Onions</td>
  </tr>
  <tr>
    <td>Celery</td>
    <td>Green onions</td>
    <td>Flour</td> <td>Salt</td><td>Black pepper</td>
  </tr>
  <tr>
    <td>Bay leaves</td>
    <td>Garlic powder</td>
    <td>Oregano</td><td>Chili flakes</td><td>Milk</td>
  </tr>
</table>
```

<!--add blurb & image by team member 4-->

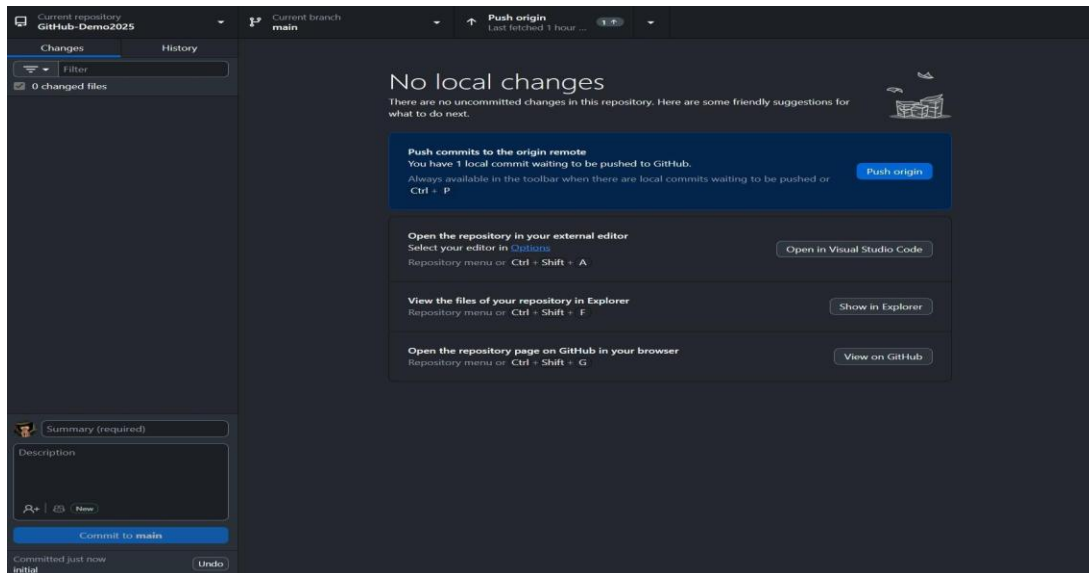
<p>Estimated costs for each dish vary by ingredients and portion size. Pizza and buffalo chicken dip are among the most affordable, costing around \$10–\$15. Pasta dishes with meat and cheese range from \$12–\$18, while teriyaki chicken averages \$12–\$16. Shrimp scampi is slightly higher due to seafood, typically \$18–\$25. Braised short ribs are the most expensive, at around \$25–\$35, largely due to the meat and wine. Overall, these meals offer a good mix of flavor and cost for home cooking.</p>

3. Once you are completely done coding your individual section, save it, and open GitHub Desktop. You will be able to see the change history now:

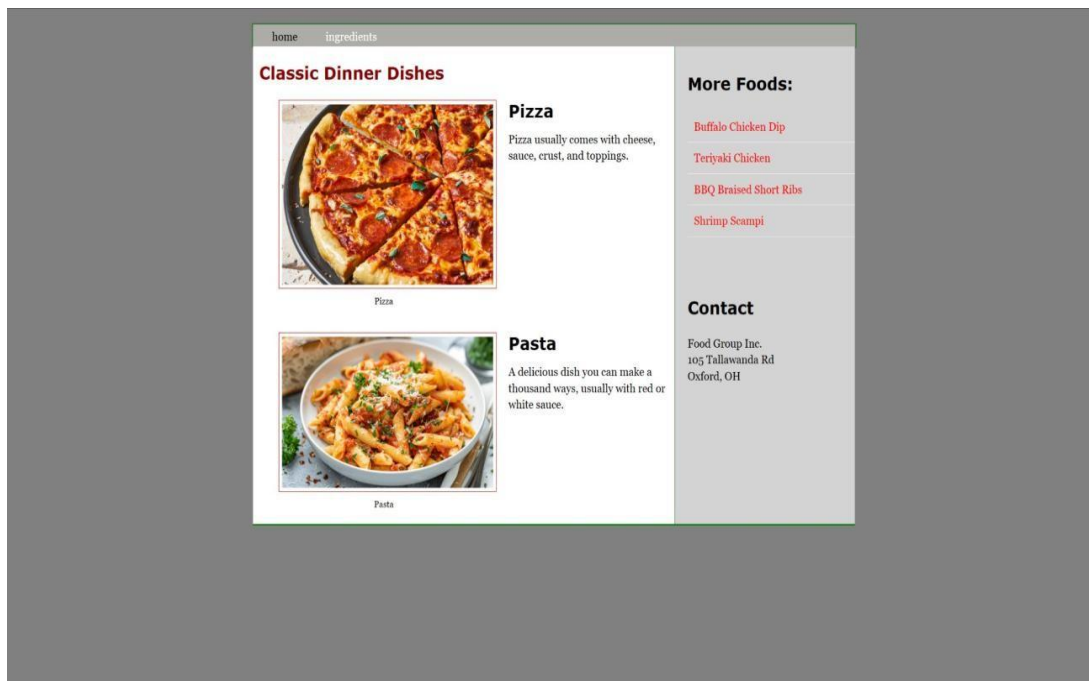


Make sure to **fetch origin and pull any changes**. Then, add a summary (e.g. updated ingredients.html) and commit files to main.

4. Once your files finish committing after adding commit description, it will ask you to push. Click “Push origin.”

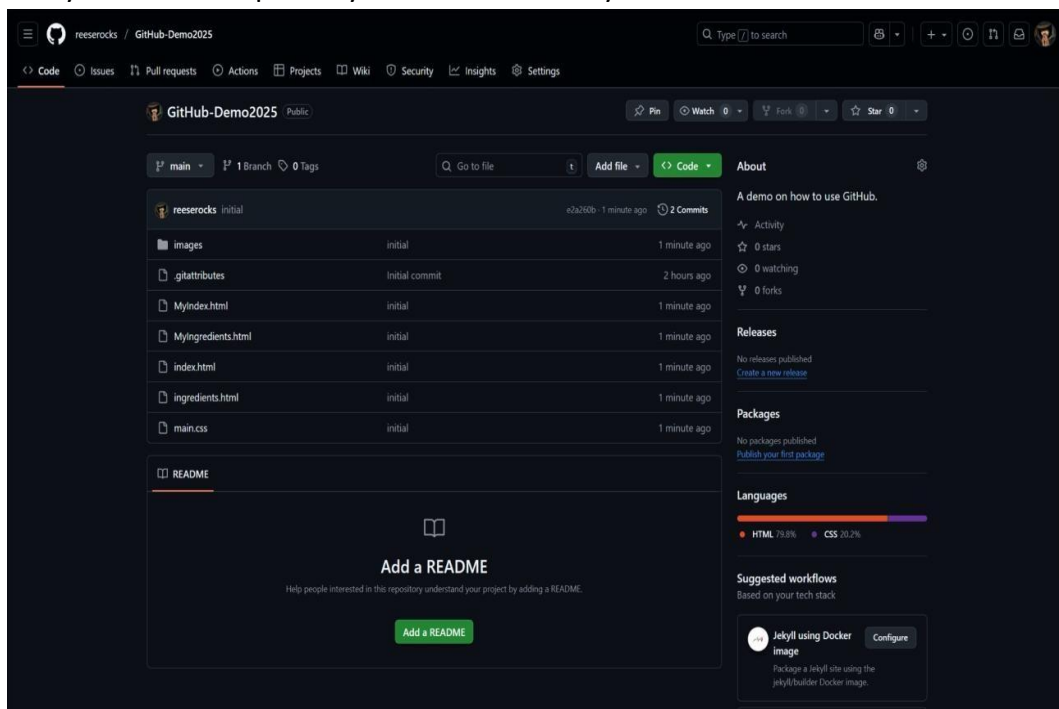


5. When you’re done, and once everyone has pushed their changes, your pages should look like this:





and your GitHub repository should include all your files:



*If you need the completed code, check out my GitHub repo [here](#).

Good work! Now you know how to use GitHub Desktop :)